NANDHA ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VIVA - VOCE

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Temporal Facial Dynamics–Driven Audio-Visual Emotion Recognition Using Vision Transformers

GUIDE:

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ABSTRACT

Emotion recognition is a key component of affective computing, enabling intelligent interaction in healthcare, education, and human-computer interaction. Many existing facial emotion recognition methods rely on static representations from a single frame, failing to capture how expressions evolve over time. This work proposes a Temporal Facial Dynamic-Driven Audio-Visual Emotion Recognition framework that uniformly samples multiple facial frames from video sequences, encodes each frame using a Vision Transformer (DINOv2), aggregates frame-level embeddings via temporal average pooling, fuses visual features with Wav2Vec 2.0 speech embeddings and uses a transformer-based fusion network for emotion classification.

EXISTING SYSTEM

Current Approach

Most existing audio-visual emotion recognition systems process facial data as static patterns — extracting features from a single video frame — completely ignoring the temporal evolution of facial expressions.

Static Frame Analysis

Only one frame per video is used, failing to capture micro-expressions and gradual emotional transitions over time.

Temporal Dynamics Ignored

Emotions evolve through subtle muscle movements; static methods miss the dynamic nature of affective states.

Misclassification Risk

Similar static appearances of emotions like 'neutral' and 'sad' lead to high confusion rates in single-frame systems.

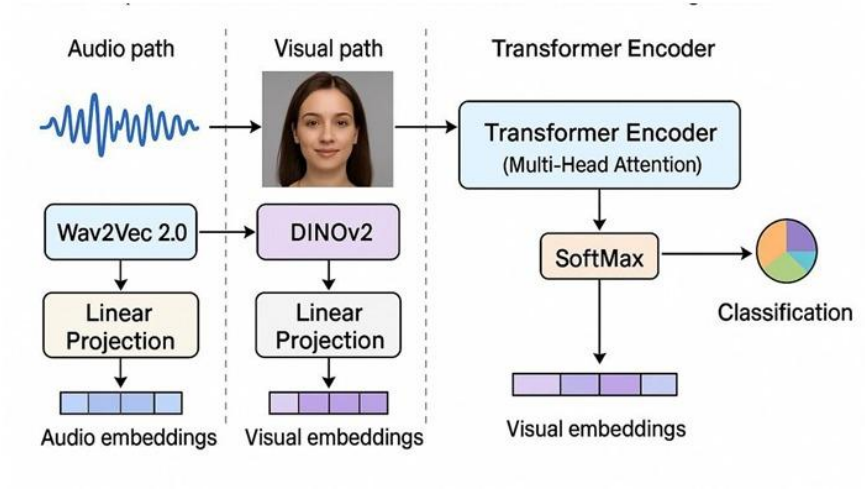
Incomplete Multimodal Fusion

Audio-visual systems without temporal facial modeling underperform; missing expression dynamics reduces overall accuracy.

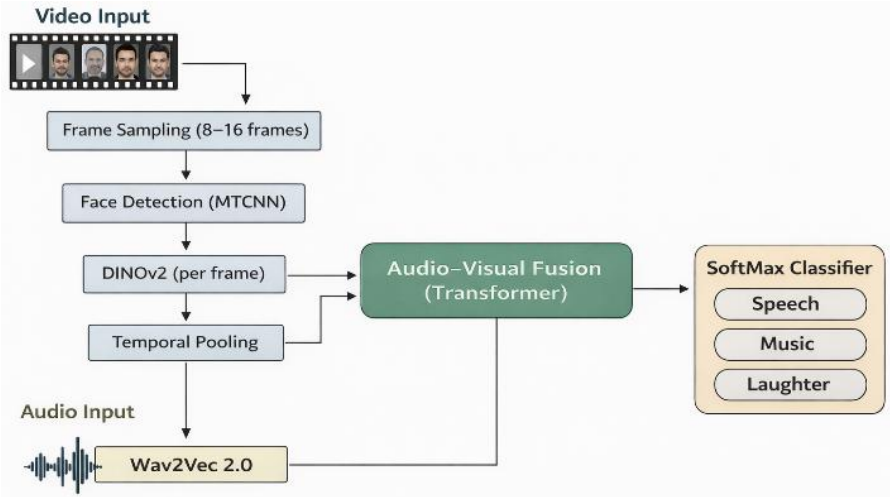
PROPOSED SYSTEM

It captures facial expression changes over time instead of relying on a single static image. Multiple frames are extracted from video to represent emotion progression. Each frame is processed using a Vision Transformer to obtain visual features. These features are combined to form a temporal representation of facial emotions and audio signals are processed using a self-supervised speech model. The system extracts important speech features like tone, pitch, and emotion cues. Audio and visual features are fused together for better understanding. A transformer-based fusion model learns relationships between both modalities. The final output is accurate emotion classification. This approach improves performance by using both temporal facial data and audio signals.

ARCHITECTURE OF PROPOSED MODEL



Architecture of Proposed wavDINO-Emotion Model



Overview of the proposed temporal facial dynamics–driven audio-visual emotion recognition framework.

LIST & DESCRIPTION MODULE

Data Preprocessing

Audio resampled to 16kHz and normalized. Facial region localized using MTCNN and resized to 224x224 px to meet Vision Transformer input requirements.

Temporal Frame Sampling

8-16 frames uniformly sampled per video. Ensures sufficient temporal coverage of facial expression dynamics across the full video duration.

Visual Feature Extraction

This module extracts facial features from video by sampling frames, detecting faces using MTCNN, and generating visual embeddings through a Vision Transformer.

Temporal Aggregation

This module combines the frame-level visual features into a single representation. Temporal average pooling is applied to capture the progression of facial expressions, helping the model understand emotion dynamics instead of static images.

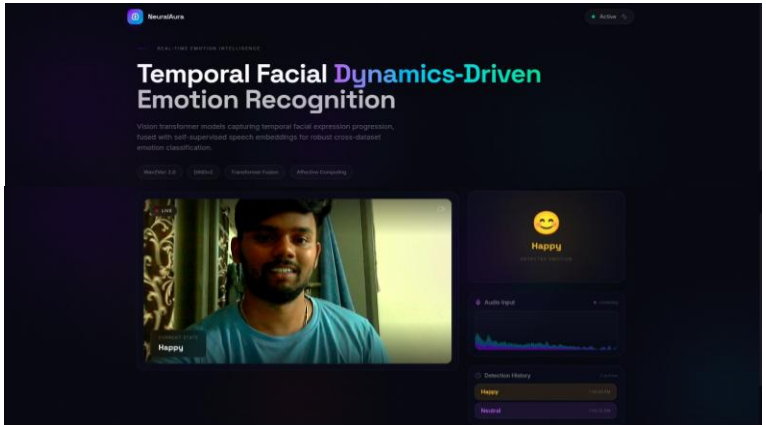
Audio Feature Extraction

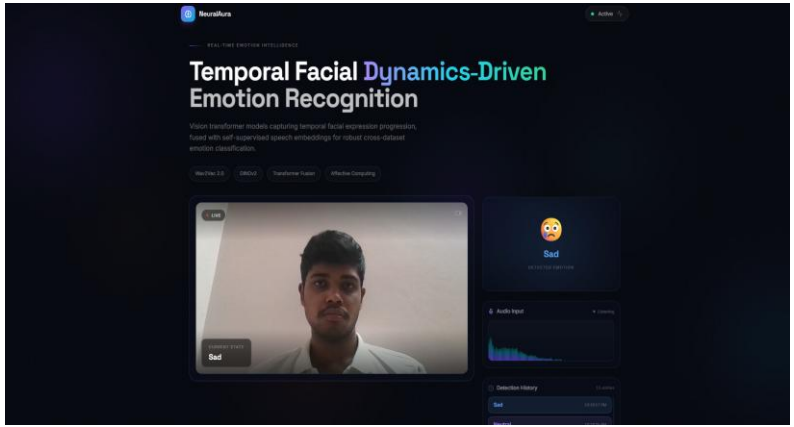
Wav2Vec 2.0 encodes raw speech waveforms into fixed-length embeddings capturing emotional acoustic properties: tone, pitch, and prosody.

Multimodal Fusion

This module merges audio and visual features using a transformer model to improve emotion recognition by learning relationships between speech and facial expressions.

SCREENSHOTS





DEMO — RESULTS & EVALUATION

Dataset-wise Performance

Dataset	Accuracy	F1-Score
CREMA-D	87.8%	0.87
RAVDESS	86.7%	0.86
AFEW	73.1%	0.70

Model Comparison

Audio-Only	84.5%	0.83
Visual – Static	83.2%	0.82
Audio-Visual – Static	86.0%	0.85
Audio-Visual – Temporal (Proposed)	**87.3%**	**0.86**

Cross-Dataset Evaluation

Train: CREMA-D => Test: RAVDESS

Accuracy: 85.0% F1-Score: 0.84

Train: CREMA-D => Test: AFEW

Accuracy: 71.4% F1-Score: 0.69

Train: RAVDESS => Test: CREMA-D

Accuracy: 83.8% F1-Score: 0.83

CODING

The complete source code for the wavDINO-Emotion framework and the Temporal Facial Dynamics-driven Audio-Visual Emotion Recognition framework, including all pre-processing pipelines, model architectures, training scripts and evaluation utilities, is available in the following GitHub repositories:

wavDINO-Emotion Framework:
https://github.com/hariprasad2005/wavDINO-Emotion

Temporal Facial Dynamics Framework:
https://github.com/hariprasad2005/temporal-audiovisual-emotion

REAL-WORLD APPLICATIONS

- ✓ ***Mental Health Monitoring:***

  AI detects emotional distress during telemedicine consultations.

- ✓ ***Customer Service Analytics:***

  Emotion detection during support calls.

- ✓ ***Smart Education Systems:***

  AI detects student confusion or engagement during online classes.

- ✓ ***Driver Monitoring Systems:***

  Detect driver fatigue or stress in vehicles.

CONCLUSION

Temporal modelling captures the evolution of facial expressions over time, enabling a better understanding of emotional dynamics. Multimodal fusion further enhances emotion recognition accuracy by combining both audio and visual features. The proposed model achieves 87.8% accuracy on the CREMA-D dataset and outperforms traditional static audio-visual models, demonstrating improved robustness, better distinction of similar emotions, and strong generalization across different datasets.

FUTURE SCOPES

- ✓ ***Lightweight Attention***

  Develop lightweight temporal attention mechanisms for faster inference without sacrificing accuracy.

- ✓ ***Real-Time Deployment***

  Engineer real-time emotion recognition pipelines suitable for live production environments.

- ✓ ***Noise Robustness***

  Improve model robustness to acoustic and visual noise in challenging real-world conditions.

- ✓ ***Telehealth & Education***

  Deploy in telehealth platforms and smart classrooms for adaptive, emotion-aware interactions.

- ✓ ***Edge Optimization***

  Optimize models for edge-device deployment (mobile, IoT) with model compression techniques.

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Thank You